

MacNeille G2/A2 Boundary on the Smallest Non-Lattice Carrier

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MacNeille G2/A2 Boundary

Pass 112 refines the MacNeille reflection search by adding finite APS A1-A4 table checks to the checker output. The result is a small but useful boundary statement: on the fixed three-element non-lattice carrier $\{0, a, b\}$ with $0 < a$ and $0 < b$, completion-created non-principal fixed cuts can coexist with finite A1-A4, and they can coexist with G2 only if A2 is dropped.

The current v1 witness `three-element-nolattice-nosynt` still has no syntactic `boxtimes` fixed point. Its MacNeille completion still has the non-principal fixed cut $\{0, a, b\}$. The new `apsAxioms` report fields show that finite A1-A4 hold for this table, but G2 fails because `boxtimes(T) <= bottom` while `T <= bottom` is false.

The exhaustive fixed-carrier search enumerated all total antitone `boxtimes` tables and all total `Box` tables on the same V-shaped carrier. It found:

package	separating tables
completion separation	216
separation + G2	54
separation + G2 + A2	0
separation + G2 + A124Core	0
separation + G2 + A1-A4 APS	0
separation + A1-A4 APS but not G2	10

Thus A2 is the first gate on this carrier. The G2 examples are vacuous-G2 examples whose antecedent is false; the finite APS examples satisfy A2 but fall on the non-G2 side.

This is not a global reflection theorem. The checker still separates finite table checks from residuals and completion-stability assumptions. The next test is to move from the fixed three-element V-carrier to four-element carriers and ask whether a genuine G2+A2 completion-separation witness appears there.

Machine artifacts:

- `artifacts/reports/pass112-macneille-g2-boundary-check.json`
- `artifacts/reports/macneille-reflection-three-element-nolattice-nosynt-v1.json`
- `code/scripts/check-pass112.py`